CHAPTER 7

Geometry identified: straight, projections and circles

Chapter overview

Objectives. Use a normal vector; calculate a project; determine or recognize a circle; translate a configuration into equations.
Prerequisites. Vector coordinates, scalar product, linear systems, square completion. Always orthonormal.
Annual benchmark. 5, 7 and 12 January: 6 a.m.

Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. $(2;-1)\cdot(1;2)$?

2. Does item (1; 2) verify 2 $x-y=0$?

3. Solve $x+y=5,\ x-y=1.$

4. Distance from (1; 2) to (4; 6) ?

5. Complete $x^2-4x.$

6. Solve x^2+y^2 = 0 in $\mathbb{R}^2.$

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Present the two conditions characterizing a project. Geometrical and algebraic evidence of a mediator.

Search before formalizing

From $P(4; 3)$ and $d : y = 2x$, compare distances to three points of d, then search and demonstrate which is the shortest.

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min. Keep a trace of a test, even if not completed. A conjecture must then be justified.

7.1 Normal Vector and Cartesian Equation

Definition and results

A normal non-zero vector at a straight line is orthogonal to any leading vector of this Right. $\vec{n} = (a; b) \neq (0; 0)$ is normal and $A(x_A; y_A)$ belongs to d, then d a for equation $a(x - x_A) + b(y - y_A) = 0$, or $ax + by + c = 0$. A possible guiding vector is ($-b; a)$.

Status and justification. $M \in d$ is equivalent to $\overrightarrow{AM} \cdot \vec{n} = 0$. In an orthonormal coordinate system, develop this product gives the equation. Conversely the set thus defined is the line passing through A orthogonal to n.

Example solved. By $A(1; 2)$ and normal $(2; -3)$: $2(x - 1) - 3(y - 2) = 0$, therefore $2x - 3y + 4 = 0$. The vector (3; 2) directs the right: produces $6 - 6 = 0$.

Point of vigilance

$(a; b)$ is normal, and not Director in general. ($a; b)$ must not be void.

Immediate application. Equation of the right by (2; $-1)$ normal (3; 4).

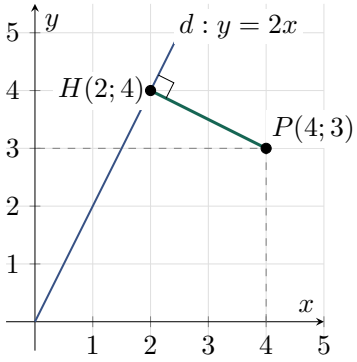
7.2 Orthogonal project and distance

Definition and results

Projected H of P over d is characterized by $H \in d$ and $\overrightarrow{PH}$ colinear to a normal vector of d. For $d : ax + by + c = 0$, write $H = P - \lambda(a; b)$, with $\lambda = \dfrac{ax_P + by_P + c}{a^2 + b^2}$. Distance is $PH = \dfrac{|ax_P + by_P + c|}{\sqrt{a^2 + b^2}}$. The distance formula is a deduced complement, not an automatism Isolated.

Status and justification. Replace $x_H = x_P - \lambda a$, $y_H = y_P - \lambda b$ in the equation: $ax_P + by_P + c - \lambda(a^2 + b^2) = 0$. The denominator is positive; H is obtained and $PH = |\lambda| \sqrt{a^2 + b^2}$.

Example solved. For $P(3; 4)$ and $d : x + y - 2 = 0$, $\lambda = \dfrac{3 + 4 - 2}{2} = \dfrac{5}{2}$, therefore $H(\dfrac{1}{2}; \dfrac{3}{2})$ and $PH = \dfrac{5}{\sqrt{2}} = \dfrac{5\sqrt{2}}{2}$. Check $x_H + y_H = 2$.



The PH segment is perpendicular to d. The two axes have the same scale; the coordinates of P and H read directly.

Point of vigilance

The distance is positive or zero: keep the absolute value. Two checks, belonging and perpendicularity, secure the project.

Immediate application. Project $P(4; 1)$ on $d : x = 2$.

7.3 Equation of a circle

Definition and results

The center circle $I(\alpha; \beta)$ and radius $r > 0$ a for equation ($(x - \alpha)^2 + (y - \beta)^2 = r^2$. One equation $x^2 + y^2 + ux + vy + w = 0$ is processed by completing the two squares. square obtained is negative, the whole is empty; if it is zero, it is a point.

Status and justification. M belongs to the circle if and only if $IM = r$. Both members being positive or null, one can raise squared and use the orthonormal distance formula.

Example solved. $x^2 + y^2 - 4x + 6y + 9 = 0$ becomes $(x - 2)^2 + (y + 3)^2 = 4$. Centre $(2; -3)$, radius 2. The signs of the centre coordinates are opposite to those in parentheses.

Point of vigilance

A quadratic equation in x and y is not always a circle: check the coefficients and the real beam.

Immediate application. Recognize $x^2 + y^2 + 2x - 8y + 17 = 0$.

7.4 Solve a configuration

Definition and results

An intersection is obtained by simultaneously solving the equations.
straight parallel to an axis, substitute the constant coordinate gives an equation of the second degree. For a mediator, write equality of distances squared; quadratic terms eliminate.

Status and justification. Equivalencies are based on belonging to both sets.
$MA = MB$ is equivalent to $MA^2 = MB^2$, since the distances are positive.

Example solved. The circle $x^2 + y^2$ = 25 and the right x = 3 cut into (3; 4) and (3; -4. The mediator of A(0; 0) and B(4; 2) check $x^2 + y^2 = (x-4)^2 + (y-2)^2$, therefore $2x + y = 5$.

Point of vigilance

An intersection is not the union of solutions sets. Check the points in both equations.

Immediate application. Intersect ($x - 1)^2 + (y + 2)^2$ = 9 with $y = 1$.

Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 7.1 · Building a Right

The plane is equipped with an orthonormal coordinate system. d go through $A(-1; 3)$ and admits the vector $\vec{n}$ = (2; 5) for normal vector.

1. Write the condition of belonging of a point $M(x; y)$ à d using a scalar product. deduce a Cartesian equation from d.
2. Provide a non-zero policy vector of d, then the coordinates of a point of d separate from A.
3. Check your point by substitution in the equation.

Exercise 7.2 · Calculate a Project

In an orthonormal coordinate system, the point is given P(4; 3) and the right d d-equation 2 $x - y$ = 0.
H the orthogonal project of P on d.

1. Determine the coordinates of H by translating the two conditions: H belongs to d and $\overrightarrow{PH}$ is normal to d.
2. Check these two conditions separately with the coordinates obtained.
3. Calculate the distance from P right d in exact form.

Exercise 7.3 · Recognize a circle

All points are considered $M(x; y)$ of the orthonormous plane whose coordinates check

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

1. Show that this set is a circle and specify its centre and radius.

2. Determine all its intersection points with the equation line $y = -2$.

3. Interpret geometrically the position of these two points on the circle.

Exercise 7.4 · Ombudsperson

In an orthonormal coordinate system, we give $A(-1; 1)$ and $B(3; 3)$. An equation of the mediator is sought. of [AB].

1. Determine this equation using the geometric properties of the mediator.
   choice of a point and a normal vector.

2. Verify that the midpoint of [AB] satisfies the equation obtained.

Exercise 7.5 · Best Access

A straight road is represented by the right d (d) equation $y = 2x$ in an orthonormal coordinate system of which unit of length is 1 km. A building is located at the point $P4; 3)$. We want to build a straight access connecting the building to a point of the road.

1. Determine the point of the road that makes this access as short as possible.
   exact access.

2. Justify that your choice is optimal across all points of the road, comparing distances
   using a right triangle. Only one observation on a drawing is not enough.

Exercise 7.6 · Parameter and Set

For a real parameter k, we define $\mathcal{E}_k$ as all points $M(x; y)$ whose coordinates check

$$x^2 + y^2 - 2x + 4y + k = 0.$$

1. Determine the nature of the $\mathcal{E}_k$ based on the value of k. Organize the response in disjointed cases.

2. When this set is a circle, specify its centre and radius; in other cases, describe
   exactly the set obtained.

Exercise 7.7 · A parabola as a locus

Optional deepening. In an orthonormal coordinate system, the point is fixed $F(0; 2)$. points $M(x; y)$ whose distance to F is equal to the distance at the axis of the abscisses.

1. Express the square of each of the two distances according to x and y.

2. Deduce an equation of the set sought. Verify the converse in order to justify this
   equation describes exactly all the points sought.

Fiscal year 7.8 · Deferred review

In an orthonormal coordinate system, $\mathcal{C}$ is the circle of diameter [AB], with $A(0; 0)$ and $B(4; 0)$.
the right d (d) equation $x = 5$.

1. Determine whether $\mathcal{C}$ and d have a common point, solving an equation.

2. Confirm the response by comparing the distance from the centre of the circle to the right and the radius of the circle circle.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 18 p. 279, questions a, b, c and d. Read a normal non-zero vector for each equation, without solving in y.

CQFD, Exercise 21 p. 279, questions a and b. Write the condition of orthogonality to normal, then check that the given point belongs to the right.

MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. Normal to 2 $x - 3y + 1 = 0$ is

A. $(3; 2)$
B. $(2; -3)$
C. $(2; 3)$
D. $(1; 0)$

2. A Director of $x + 2y = 0$ is

A. $(-2; 1)$
B. $(1; 2)$
C. $(2; 1)$
D. $(0; 2)$

3. Centre of $x + 2)^2 + (y - 3)^2 = 4$:

A. $(2; 3)$
B. $(-2; -3)$
C. $(3; -2)$
D. $(-2; 3)$

4. Rayon of $x^2 + y^2 = 9$:

A. 9
B. 81
C. 3
D. $\sqrt{3}$

5. $x^2 + y^2 = -1$ described

A. a circle
B. Empty assembly
C. Origin
D. a right

MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. Projected from (3; 4) on the abscissa axis:

A. $(0; 4)$
B. $(0; 0)$
C. $(3; 0)$
D. $(4; 3)$

2. Distance from (1; 5) to $y = 2$:

A. 3
B. 5
C. 1

D. $\sqrt{10}$

3. To define $ax + by + c = 0$ as right, it is necessary

A. $a \neq 0$ only

B. $b \neq 0$ only

C. $c \neq 0$

D. $(a; b) \neq (0; 0)$

4. $x^2 + y^2 = 0$ described

A. A circle of radius 1

B. the only point O

C. a right

D. The plan

5. Circle $x^2 + y^2 = 4$ and straight $x = 2$: number of commonalities

A. 0

B. 2

C. 1

D. Infinite

True or false?

1. The orthogonal project of P on a right is unique.

2. Any equation $x^2 + y^2 + c = 0$ describes a positive radius circle.

3. A right may encounter a circle in three distinct points.

Personal review

I give a nonzero normal; I check the projected; I complete the two squares; I distinguish circle, point and empty set.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.

Corrected and conduct of chapter 7

Use as a teacher. The following answers relate to the outstanding tasks in the course.
first clarify the approach; provide graduated help before the solution.
estimates to be calibrated in class.

Diagnosis and Remediation

1. 0. *Action:* Take over the criterion of orthogonality.

2. Yes: 2 $-\ 2 = 0$. *Action:* Replace x and y simultaneously.

3. $x = 3,\ y = 2$. *Action:* Add and replace.

4. 5. *Action:* Calculate the vector and then its norm.

5. $(x-2)^2 - 4$. *Action:* Develop to check.

6. The only torque (0; 0). *Action:* Use the sum of two positive squares.

Research activity

Trials suggest H(2;4), but Pythagoras in PHM demonstrates minimality.
with normal (2;-1).

Immediate applications

1. $3(x-2) + 4(y+1) = 0$, or 3 $x + 4y - 2 = 0$.

2. H(2; 1) and distance 2; normal is horizontal.

3. $(x+1)^2 + (y-4)^2 = 0$: the only point (-1; 4), not a strictly positive radius circle.

4. $(x-1)^2 = 0$, so only point (1; 1); the right is tangent to the circle.

Exercises: approach and results

Exercise 7.1 · Build a straight line. $2(x+1) + 5(y-3) = 0$, or 2 $x + 5y - 13 = 0$.
$(5; -2)$, therefore B(4; 1) is on the right; control 8 + 5 $-\ 13 = 0$.

Exercise 7.2 · Calculate a project. Normal (2; $-1)$; $\lambda = \dfrac{8-3}{5} = 1$. $H = (4;3) - (2;-1) = (2;4)$. $2(2) - 4 = 0$ and $\overrightarrow{PH} = (-2;\ 1)$ is normal; distance $\sqrt{5}$.

Exercise 7.3 · Recognize a circle. $(x-3)^2 + (y+2)^2 = 25$; centre (3; $-(2)$, section 5.
$y = -2,\ x = 3 \pm 5$: $(-2; -(2)$ and $(8; -2)$.

Exercise 7.4 · Mediator. Environment I(1;2), normal $\overrightarrow{AB} = (4;2)$. $4(x-1) + 2(y-2) = 0$, or $2x + y - 4 = 0$. I check 2 + 2 $-\ 4 = 0$.

Exercise 7.5 · The best access. Projected H(2;4) minimizes PH, length $\sqrt{5}$ km. For all M on the road, the PHM triangle is right-angled at H and $PM^2 = PH^2 + HM^2 \geq PH^2$. Equality for M=H.

Exercise 7.6 · Parameter and set. $(x-1)^2+(y+2)^2=5-k$. If $k<5$, center circle $(1;-2)$ and radius $\sqrt{5-k}$; if $k=5$, point; if $k>$ Five, empty.

Exercise 7.7 · A parabola as a locus. $MF^2=x^2+(y-2)^2$, distance to square axis y^2. Equality: $x^2-4y+4=0$, or $y=\frac{x^2}{4}+1$. Positive distances allow for mutual exchange.

Exercise 7.8 · Deferred review. Centre (2; 0), radius 2 ; ($x-2)^2+y^2=4$. For x=5, $y^2=-5$, Impossible. Interpretation: Right at distance 3 from the centre, above radius.

CQFD, Exercise 18 p. 279, questions a, b, c and d. Possible responses are ($-1;2)$, $(3;-6)$, $(\frac{1}{4};-1)$ and (1; 0). Any non-zero multiple is suitable. $x=4$, write $x-4=0$.

CQFD, Exercise 21 p. 279, questions a and b. a : $x-(y-2)=0$, or $x-y+2=0$. b : $4(x+1)+6(y-3)=0$, or 2 $x+3y-7=0$. Substitute coordinates of points confirms membership.

MCQ: answers and reasons

MCQ 1.

1. B. X and y coefficients.
2. A. Product with normal zero.
3. D. Read the coordinates opposite the signs.
4. C. Positive root of 9.
5. B. Sum of non-negative squares.

MCQ 2.

1. C. Same abscissa, ordered null.
2. A. Absolute vertical deviation.
3. D. Non-zero normal.
4. B. The two squares shall be void.
5. C. $y^2=0$.

True or false

1. True: per P passes a single perpendicular to the right; if P belongs to the right, H=P.
2. False: if c=0, one dot; if c>0, empty set.
3. False: substitution of a right parameter in the circle equation gives an equation quadratic with strictly positive dominant coefficient, therefore at most two roots.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.